VLSI Task 6: Rectenna Monitoring and Load Control

Verilog RTL and Testbench — Xilinx Vivado

Design source: **rectenna_load_control.v**. Simulation source: **tb_rectenna_load_control.v**.

1. Verilog RTL Code

```
module rectenna_load_control (
    input clk,
    input rst_n,
    input [7:0] power_level,
    input [7:0] power_threshold,
    output reg load_enable,
    output reg power_status
);

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            load_enable  <= 1'b0;
            power_status <= 1'b0;
        end
        else begin
            if (power_level >= power_threshold) begin
                power_status <= 1'b1;
                load_enable  <= 1'b1;
            end
            else begin
                power_status <= 1'b0;
                load_enable  <= 1'b0;
            end
        end
    end

endmodule
```

2. Verilog Testbench Code

```
`timescale 1ns / 1ps

module tb_rectenna_load_control;

    reg clk = 0;
    reg rst_n = 0;
    reg [7:0] power_level = 0;
    reg [7:0] power_threshold = 100;

    wire load_enable;
    wire power_status;

    rectenna_load_control uut (
        .clk(clk),
        .rst_n(rst_n),
        .power_level(power_level),
        .power_threshold(power_threshold),
        .load_enable(load_enable),
        .power_status(power_status)
    );

    always #5 clk = ~clk;

    initial begin
        #20;
        rst_n = 1;

        // Test 1: Insufficient power
        @(negedge clk);
        power_level = 8'd50;
        @(negedge clk);

        // Test 2: Power reaches threshold
        power_level = 8'd100;
        @(negedge clk);

        // Test 3: Power exceeds threshold
        power_level = 8'd150;
        @(negedge clk);

        // Test 4: Power drops below threshold
        power_level = 8'd70;
        @(negedge clk);

        // Test 5: Change threshold to 60
        power_threshold = 8'd60;
        @(negedge clk);

        #20;
        $finish;
    end

endmodule
```

I/O definition

Inputs: clk, rst_n, power_level[7:0], power_threshold[7:0]. Outputs: load_enable and power_status.

Expected behavior

Power level 50 with threshold 100: load OFF. Power 100 or 150 with threshold 100: load ON. Power 70 with threshold 100: load OFF. After lowering threshold to 60, power 70 enables the load.

Note

This is a digital monitoring/control simulation. It does not model RF rectification or guarantee sufficient real harvested energy; actual hardware needs sensor calibration and suitable power/load protection.